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Student Number



Barker
College

2025

**TRIAL HIGHER SCHOOL
CERTIFICATE EXAMINATION**

Mathematics Extension 2

Staff Involved:

PM MONDAY 11TH AUGUST

- GDH - Mr Hanlon
- BHC - Mr Carruthers*
- ALY - Ms Young

50 copies

Topics Covered

TOPICS COVERED		
Complex Numbers	Proof	Integration
Vectors	Mechanics	

General

Instructions:

- Write your Student Number at the top of this page and on all writing booklets
- Reading time - 10 minutes
- Working time - 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A separate reference booklet is provided
- Marks may not be awarded for careless or poorly arranged working
- Diagrams are not to scale
- For questions in Section II, show relevant mathematical reasoning and / or calculations

Total marks:

100

Section I - 10 marks (pages 2 - 5)

- Attempt Questions 1 - 10
- Allow about 20 minutes for this section

Section II - 90 marks (pages 7 - 14)

- Attempt Questions 11 - 16
- Allow about 2 hours and 40 minutes for this section
- Answer each question in a SEPARATE writing booklet.

Section I

10 marks

Attempt Questions 1 – 10

Allow about 20 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 10.

- 1 The argument of a complex number z is $\frac{\pi}{7}$.

What is $\text{Arg}(z^{10})$ equal to?

- A. $\frac{3\pi}{7}$
- B. $\frac{-3\pi}{7}$
- C. $\frac{4\pi}{7}$
- D. $\frac{-4\pi}{7}$

- 2 Let $\underline{a} = \begin{pmatrix} -4 \\ -2 \\ 6 \end{pmatrix}$ and $\underline{b} = \begin{pmatrix} -2m \\ 2 \\ 2 \end{pmatrix}$, where m is a real constant. For what value of m will the vector $\underline{b} - \underline{a}$ be perpendicular to the vector $\underline{b}$?

- A. 0 only
- B. 2 only
- C. 0 or 2
- D. 0 or -2

3 Consider the following proposition.

“If a positive integer is not a perfect square, then it is not prime.”

If a number was found which had the following properties, which would be a counterexample to the proposition?

- A. A positive integer that is not a perfect square and is prime.
- B. A positive integer that is not a perfect square and is not prime.
- C. A positive integer that is a perfect square and is prime.
- D. A positive integer that is a perfect square and is not prime.

4 A particle moves along a curve so that its position at time t is given by $\vec{x} = \begin{pmatrix} 2t \\ t^2 \\ t^3 \end{pmatrix}$.

The acceleration at $t = 1$ is:

- A. $2\vec{j} + 6\vec{k}$
- B. $\vec{i} + \vec{j} + 6\vec{k}$
- C. $\vec{i} + 2\vec{j} + 6\vec{k}$
- D. $\vec{j} + 6\vec{k}$

5 Which of the following is an expression for $\int \frac{\sin^3 x - \cos^3 x}{\sin x - \cos x} dx$?

- A. $x - \frac{1}{2} \sin^2 x + c$
- B. $x + \frac{1}{2} \sin^2 x + c$
- C. $x + \frac{1}{2} \cos 2x + c$
- D. $x - \frac{1}{2} \cos 2x + c$

6 Which of the following statements is the negation of the statement

“ $\forall x \in \mathbb{R}^+, \exists y \in \mathbb{R}^+$ such that $xy = 1$ ” ?

A. $\exists x, y \in \mathbb{R}^+$ such that $xy \neq 1$

B. $\exists x \in \mathbb{R}^+$ such that $\forall y \in \mathbb{R}^+, xy \neq 1$

C. $\forall x, y \in \mathbb{R}^+, xy \neq 1$

D. $\forall x \in \mathbb{R}^+$, there exists a positive real number y such that $xy \neq 1$

7 Using the substitution $x = \pi - u$, the definite integral $\int_0^\pi x \sin x \, dx$ will simplify to:

A. 0

B. $\frac{\pi^2}{4} \int_0^\pi \sin x \, dx$

C. $\frac{\pi}{2} \int_0^\pi \sin x \, dx$

D. $\int_0^\pi \sin x \, dx$

8 Let z be a complex number such that $z^3 = \frac{i}{\bar{z}}$.

Which of the following is a possible value for z ?

A. $e^{i\frac{\pi}{4}}$

B. $e^{-i\frac{\pi}{4}}$

C. $e^{i\frac{\pi}{3}}$

D. $e^{-i\frac{\pi}{3}}$

9 Which of the following statements is the contrapositive of the statement

“if $x > y > 0$ then $0 < \frac{1}{x} < \frac{1}{y}$ ”?

A. If $0 \geq \frac{1}{x}$ or $\frac{1}{x} \geq \frac{1}{y}$ then $x \leq y$ or $y \leq 0$.

B. If $0 \geq \frac{1}{x}$ and $\frac{1}{x} \geq \frac{1}{y}$ then $x \leq y$ and $y \leq 0$.

C. If $0 \geq \frac{1}{x} \geq \frac{1}{y}$ then $x \leq y \leq 0$.

D. If $x \leq y \leq 0$ then $0 \geq \frac{1}{x} \geq \frac{1}{y}$.

10 Which of the following integrals has the largest value?

A. $\int_0^{\frac{\pi}{4}} \sin x \, dx$

B. $\int_0^{\frac{\pi}{4}} \sin^2 x \, dx$

C. $\int_0^{\frac{\pi}{4}} 1 - \sin x \, dx$

D. $\int_0^{\frac{\pi}{4}} 1 - \sin^2 x \, dx$

End of Section I

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Section II

90 marks

Attempt Questions 11 - 16

Allow about 2 hours and 40 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11 - 16 your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) [START THE QUESTION 11 WRITING BOOKLET]

(a) Let $z = a - 2bi$. Find complex numbers for the following, in the form $x + iy$.

(i) $\frac{1}{\bar{z}}$ 2

(ii) z^2 1

(b) (i) Using Euler's formula, $e^{i\theta} = \cos \theta + i \sin \theta$, find an expression for $e^{-i\theta}$ in terms of $\cos \theta$ and $\sin \theta$. 1

(ii) Hence show that $\sin \theta = \frac{1}{2i}(e^{i\theta} - e^{-i\theta})$. 1

(iii) Use the result from part (ii) to show that $\sin^3 \theta = \frac{3}{4} \sin \theta - \frac{1}{4} \sin 3\theta$. 3

(c) (i) Find real numbers A and B such that $\frac{6x^2 + 3x + 17}{(x-3)(x^2 + 2x + 5)} = \frac{A}{x-3} + \frac{Bx+1}{x^2 + 2x + 5}$. 2

(ii) Hence find $\int \frac{6x^2 + 3x + 17}{(x-3)(x^2 + 2x + 5)} dx$. 3

(d) Prove the following statement for positive n using a proof by contradiction.

“For each irrational number n , $\sqrt{n}$ is also irrational.” 2

Question 12 (15 marks)**[START THE QUESTION 12 WRITING BOOKLET]**

- (a) Assume that the tides rise and fall in simple harmonic motion. At low tide the depth of water in a harbour was 6 m deep, and at high tide it was 10 m deep. Low tide was at 8:00 am and the high tide following it was at 3:00 pm.

(i) What is the amplitude of the motion? 1

(ii) What is the period of the motion? 1

(iii) What was the depth of the harbour at 11:30 am? 1

- (iv) Starting with $y = a \cos(nt + \alpha) + c$ and by using your answers from parts (i), (ii) and (iii), show that the depth y m of the water in the harbour is given by

$$y = 8 - 2 \cos\left(\frac{\pi}{7}t\right) \text{ where } t \text{ is the number of hours after 8:00 am.} \quad 2$$

- (v) A boat needs at least 9 m of water to pass through the harbour. Between which times after 8:00 am on this day could the ship navigate safely through the harbour? 2

- (b) $P(z) = z^4 - 10z^3 + 117z^2 - 412z + 1460$.

$P(z)$ has zeroes $2a + bi$ and $3a + 2bi$, where a and b are integers.

(i) Find the values of a and b and hence find all the zeroes of $P(z)$. 3

(ii) Factorise $P(z)$ into its quadratic factors with real coefficients. 1

- (c) It is given that $3^x > 2x + \frac{1}{2} \ln x$ for $x > 0$.

By forming a series of n inequalities, show that $3^{n+1} > 2n^2 + 2n + 3 + \ln(n!)$. 4

Question 13 (15 marks)**[START THE QUESTION 13 WRITING BOOKLET]**

- (a) Given that $u_1 = 1$ and $u_{n+1} = 2u_n + (-1)^{n+1}$, prove that $u_n = \frac{1}{3}(2^{n+1} + (-1)^n)$, where $n \in \mathbb{Z}^+$, by mathematical induction. **3**
- (b) (i) Use a suitable substitution to show that $\int_0^a f(x) \, dx = \int_0^a f(a-x) \, dx$. **2**
- (ii) Hence, or otherwise, evaluate $\int_0^{\frac{\pi}{4}} \frac{\sin 2x}{\sin 2x + \cos 2x} \, dx$. **4**
- (c) The acceleration of a particle P moving along the x -axis is given by $\ddot{x} = 4x^3 - 12x$.
- (i) If P is initially at the origin with velocity u , show that its velocity is given by $v^2 = 2x^4 - 12x^2 + u^2$. **2**
- (ii) If $u = 4$, show that the particle oscillates within the interval $-\sqrt{2} \leq x \leq \sqrt{2}$. **3**
- (iii) Is the motion referred to in part (ii) simple harmonic motion? **1**
Give a reason for your answer.

Question 14 (15 marks)**[START THE QUESTION 14 WRITING BOOKLET]**

- (a) When an aircraft touches down on a runway, two different retarding forces combine to ultimately bring the aircraft to rest. There is a constant frictional force of $\frac{-1}{5}M$ newtons, where the mass of the aircraft is M kg. There is also a force of $\frac{-1}{125}Mv^2$ newtons, where v is the aircraft's velocity in m/s, due to the reverse thrust of the engines. The reverse thrust of the engines is applied from 10 seconds after touchdown until the plane stops.
- (i) The aircraft's speed at touchdown is 80 m/s.
Show that, at the instant the reverse thrust of the engines takes effect, $v = 78$ m/s and the aircraft's displacement from its touchdown point $x = 790$ m. **3**
- (ii) Show that for $t > 10$, and until the aircraft stops, $\frac{d^2x}{dt^2} = \frac{-1}{125}(25 + v^2)$. **1**
- (iii) Show that when $t > 10$, $x = 790 + 62.5[\ln 6109 - \ln(25 + v^2)]$. **3**
- (iv) Calculate how far from the touchdown point the aircraft travels before it comes to rest. Answer to the nearest metre. **1**
- (b) Three different particles move in the xy -plane according to trajectories defined parametrically by:
- $$\underline{R}_1(t) = \cos(t)\underline{i} + \sin(t)\underline{j}$$
- $$\underline{R}_2(t) = \cos(2t)\underline{i} + \sin(2t)\underline{j}$$
- $$\underline{R}_3(t) = \sin(t)\underline{i} + \cos(t)\underline{j}$$
- For each particle, describe:
1. the shape of the path
 2. the period of the motion
 3. the direction of the motion
- 3**
- (c) A body of mass 1 kg is moving in a horizontal straight line and is acted on by a force $F = 8x^3 + 36x$ newtons, where x is the displacement of the body in metres, measured from the origin.
- If the body is initially one metre to the right of the origin with velocity of 11 m/s, find the displacement function $x(t)$. **4**

Question 15 (15 marks)**[START THE QUESTION 15 WRITING BOOKLET]**

(a) There are two lines $R_1 = \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$ and $R_2 = \begin{pmatrix} 3 \\ 0 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}$.

(i) Show that these lines intersect and find the point of intersection. 2

(ii) Find the shortest distance from the point $P(1, 4, 0)$ to the line R_1 . 3

(b) Let $I_n = \int_0^1 \frac{1}{(1+x^2)^n} dx$ for $n \geq 1$.

(i) Use integration by parts to show that $I_n = \frac{1}{2^n} + 2n \int_0^1 \frac{x^2}{(1+x^2)^{n+1}} dx$. 2

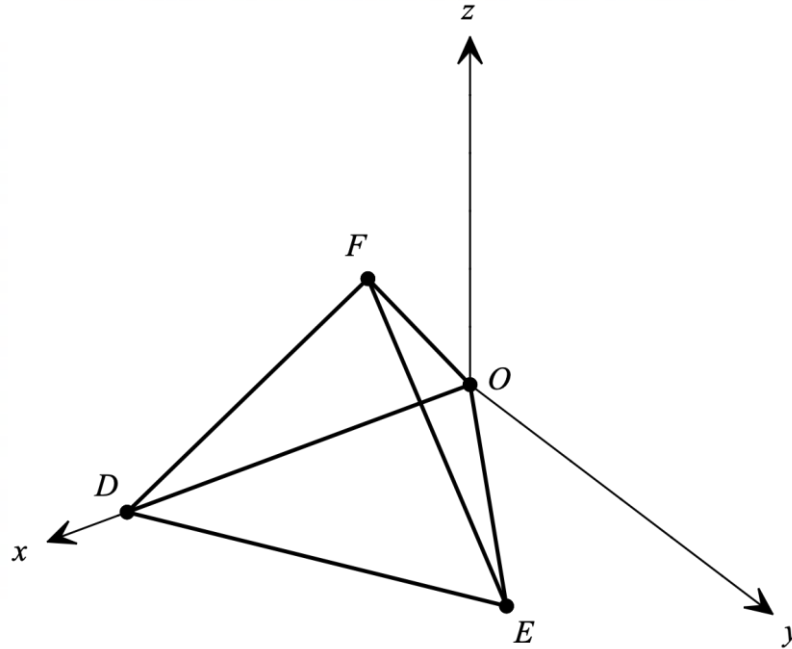
(ii) Hence, or otherwise, show that $I_{n+1} = \frac{1}{n2^{n+1}} + \frac{2n-1}{2n} I_n$. 3

(iii) Evaluate $\int_0^1 \frac{1}{(1+x^2)^2} dx$. 1

Question 15 continues on page 12

Question 15 (continued)

- (c) The faces of the tetrahedron $ODEF$ are equilateral triangles of side length 1 unit. The base ODE lies in the xy -plane with one vertex at $O(0,0,0)$ and one vertex at $D(1,0,0)$, with F above the xy -plane



- (i) Find the coordinates of E . **1**
- (ii) Find the coordinates of F . **3**

End of Question 15

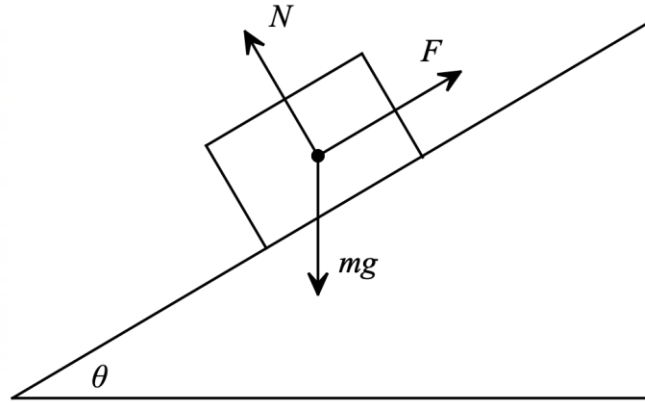
Question 16 (15 marks)**[START THE QUESTION 16 WRITING BOOKLET]**

- (a) Find $\int \frac{dx}{\sqrt{4x^2 - 1}}$ by making a suitable substitution. **3**
- (b) Given that the product of n consecutive numbers is divisible by $n!$, use mathematical induction to prove that $n^5 - n$ is divisible by 2, 3 and 5 for integers $n \geq 2$. **4**
- (c) Consider the line ℓ defined parametrically by $\underline{R}(\lambda) = \underline{a} + \lambda \underline{b}$ and the sphere S defined by $|\underline{y} - \underline{c}| = r$.
- (i) Suppose ℓ is a tangent to S . Given that the line and the sphere have only one point of intersection, prove that the position vector of the point of contact is given by $\underline{R}(\lambda)$ when $\lambda = \frac{\underline{b} \cdot (\underline{c} - \underline{a})}{|\underline{b}|^2}$. **3**
- Note: you may **not** use the fact that the tangent is perpendicular to the radius.*
- (ii) Hence, show that the tangent to the sphere is perpendicular to the radius at the point of contact. **1**

Question 16 continues on page 14

Question 16 (continued)

- (d) An object of mass m rests on a plane inclined at a variable angle θ from the horizontal. It experiences a friction force, F , and a normal force, N . The friction force is $F = \mu N$ where μ is constant. The normal force, N , is equal in magnitude to the force the object exerts on the plane perpendicular to the plane. Let α be the maximal angle that the plane can be tilted before the object begins to slide down the plane.



For some $\theta > \alpha$, a horizontal force H is applied to the object so that if H is sufficiently large, then the object will slide up the plane and if H is sufficiently small, then the object will slide down the plane.

Show that, for the object to remain motionless, H must satisfy

$$mg \tan(\theta - \alpha) \leq H \leq mg \tan(\theta + \alpha)$$

4

End of Paper

2025 Trial Extension 2

① $z^{10} = (r \cos \frac{\pi}{7})^{10}$
 $= r^{10} \cos \frac{10\pi}{7}$
 $\therefore \text{Arg}(z^{10}) = \frac{10\pi}{7} = -\frac{4\pi}{7} \therefore \boxed{D}$

② $b-a = \begin{pmatrix} -2m+4 \\ 4 \\ -4 \end{pmatrix}$
 $(b-a) \cdot b = 0$ for perpendicular
 $\therefore \begin{pmatrix} -2m+4 \\ 4 \\ -4 \end{pmatrix} \cdot \begin{pmatrix} -2m \\ 2 \\ 2 \end{pmatrix} = 0$
 $\therefore 4m^2 - 8m + 8 - 8 = 0$
 $4m(m-2) = 0$
 $m=0$ or $2 \therefore \boxed{C}$

③ Need a true integer that is not a p.f. $\square$
 but is prime $\therefore \boxed{A}$

④ $\dot{x} = \begin{pmatrix} 2 \\ 2t \\ 3t^2 \end{pmatrix} \ddot{x} = \begin{pmatrix} 0 \\ 2 \\ 6t \end{pmatrix}$
 At $t=1, \ddot{x} = \begin{pmatrix} 0 \\ 2 \\ 6 \end{pmatrix} \therefore \boxed{A}$

⑤ $\int \frac{(\sin x - \cos x)(\sin^2 x + \sin x \cos x + \cos^2 x)}{(\sin x - \cos x)} dx$
 $= \int \sin^2 x + \cos^2 x + \sin x \cos x dx$
 $= \int 1 + \frac{1}{2} \sin 2x dx$
 $= x + \frac{\sin^2 x}{2} + C \therefore \boxed{B}$

⑥ $\exists x \in \mathbb{R}^+, \forall y \in \mathbb{R}^+, xy \neq 1 \therefore \boxed{B}$
 The original statement basically says:
 'All true real x have a reciprocal and the reciprocal is also true real'.
 The negation therefore is:
 'There exists a true real x which does not have a true real reciprocal'.
 Note that the original statement is true & thus the negation should be false.

⑦ $x=0, u=\pi$
 $x=\pi, u=0$

$\frac{dx}{du} = -1 \therefore dx = -du$

$I = \int_{\pi}^0 (\pi-u) \sin(\pi-u) (-du)$
 $= \int_0^{\pi} (\pi-u) \sin u du = \pi \int_0^{\pi} \sin u du - \int_0^{\pi} u \sin u du$

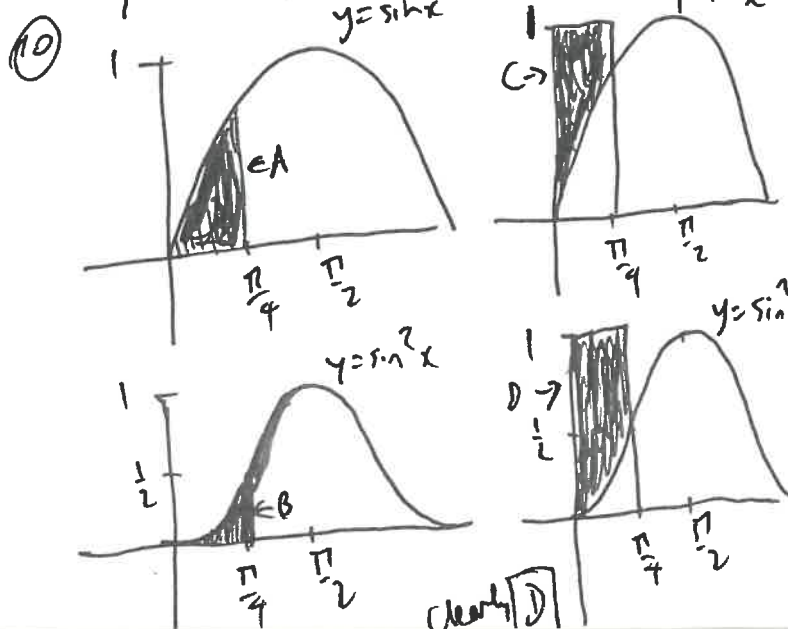
$\therefore I = \int_0^{\pi} x \sin x = \pi \int_0^{\pi} \sin u du - \int_0^{\pi} x \sin x dx$
 via change of variable

$\therefore I = \int_0^{\pi} x \sin x = \pi \int_0^{\pi} \sin x dx - I$
 $\therefore 2I = \pi \int_0^{\pi} \sin x dx \therefore I = \frac{\pi}{2} \int_0^{\pi} \sin x dx$
 $\therefore \boxed{C}$

⑧ $z^3 \bar{z} = i$
 $A: e^{i\frac{3\pi}{4}} e^{i(\frac{\pi}{4})} = e^{i\frac{2\pi}{4}} = e^{i\frac{\pi}{2}} = i$
 $\cos \frac{\pi}{2} + i \sin \frac{\pi}{2} = i \therefore \boxed{A}$
 [similar for B, C, D but none work]

⑨ $\sim (0 < \frac{1}{x} < \frac{1}{y}) \rightarrow \sim (x > y > 0)$

$\therefore$ If $0 > \frac{1}{x}$ or $\frac{1}{x} \geq \frac{1}{y}$ then
 $x \leq y$ or $y \leq 0 \therefore \boxed{A}$
 Note that to negate 'AND' we reverse inequalities and change to 'OR'.
 A Squeezed inequality is an AND situation.



(1) (a) (i) $\bar{z} = a + 2bi$

(2) $\frac{1}{\bar{z}} = \frac{1}{a+2bi} \times \frac{a-2bi}{a-2bi}$
 $= \frac{a-2bi}{a^2+4b^2} = \frac{a}{a^2+4b^2} - \frac{2b}{a^2+4b^2}i$

(ii) $z^2 = a^2 - 4abi - 4b^2$

(1) $= a^2 - 4b^2 - 4abi$

b) (i) $e^{-i\theta} = \cos(-\theta) + i\sin(-\theta)$

(1) $= \cos\theta - i\sin\theta$

(ii) Subtract vertically:
 $e^{i\theta} - e^{-i\theta} = 2i\sin\theta$

(1) $\therefore \sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$

(iii) $\sin^3\theta = \frac{e^{3i\theta} - 3e^{i\theta} + 3e^{-i\theta} - e^{-3i\theta}}{8i^3}$

(2) $= \frac{e^{3i\theta} - e^{-3i\theta} - 3(e^{i\theta} - e^{-i\theta})}{-8i}$

$= \frac{2i\sin 3\theta - 3 \times 2i\sin\theta}{-8i}$

$= \frac{2\sin 3\theta - 6\sin\theta}{-8}$

$= \frac{\sin 3\theta - 3\sin\theta}{-4}$

$= \frac{3\sin\theta - \sin 3\theta}{4}$

$= \frac{3}{4}\sin\theta - \frac{\sin 3\theta}{4}$

(c) (i) $6x^2 + 3x + 17 = A(x^2 + 2x + 5) + (Bx + 1)(x - 3)$

let $x = 3$: $80 = 20A \therefore A = 4$ (2)

Coeff x^2 : $6 = 4 + B \therefore B = 2$

(ii) $\int \frac{4}{x-3} + \frac{2x+1}{x^2+2x+5} dx$ (3)

$= 4\ln|x-3| + \int \frac{2x+1}{x^2+2x+5} \cdot \frac{1}{(x+1)^2+4} dx$

$= 4\ln|x-3| + \ln|x^2+2x+5| - \frac{1}{2} \tan^{-1}\left(\frac{x+1}{2}\right) + C$

(d) let n be irrational & $\sqrt{n}$ rational
 $\therefore \sqrt{n} = \frac{a}{b}, a, b \in \mathbb{Z}^+$

$\therefore n = \frac{a^2}{b^2}$

$\therefore n$ rational
 But n irrational! $\therefore$ Contradiction
 $\therefore$ statement is true. ✓

(12) (i) amp = $\frac{10-6}{2} = 2m$

(ii) period = 7 hours
 $\therefore$ period = 14 hours

(iii) $3\frac{1}{2}$ hours is halfway between low & high tide
 $\therefore$ Centre of motion $\therefore 8m$

(iv) $y = 2\cos\left(\frac{\pi}{7}t + \alpha\right) + 8$
 Period = $14 = \frac{2\pi}{\frac{\pi}{7}} \therefore \frac{\pi}{7} = \frac{\pi}{T}$

$\therefore y = 2\cos\left(\frac{\pi}{7}t + \alpha\right) + 8$

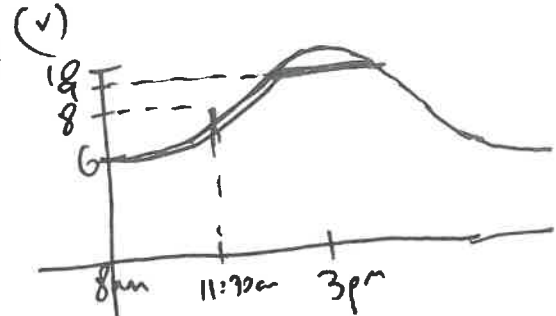
$y = 2\cos\frac{\pi}{7}t \cos\alpha - 2\sin\frac{\pi}{7}t \sin\alpha + 8$

when $t = 0, y = 6$

$\therefore 6 = 2\cos\alpha + 8 \therefore \cos\alpha = -1$
 $\therefore \alpha = \pi$

$\therefore y = 2\cos\frac{\pi}{7}t \cos\pi - 2\sin\frac{\pi}{7}t \sin\pi + 8$

$y = 2\cos\frac{\pi}{7}t(-1) + 8 = 8 - 2\cos\left(\frac{\pi}{7}t\right)$



$q = 8 - 2\cos\left(\frac{\pi}{7}t\right)$

$\cos\left(\frac{\pi}{7}t\right) = -\frac{1}{2}$

$\therefore \frac{\pi}{7}t = \frac{2\pi}{3}, \frac{4\pi}{3}$

$t = \frac{14}{3}, \frac{28}{3} \therefore 4 \text{ hr } 40 \text{ min, } 9 \text{ hr } 20 \text{ min}$

$\therefore 12:40 \text{ pm} - 5:20 \text{ pm}$

12 (b)(i) since real coefficients,
zeros: $2a \pm bi$, $3a \pm 2bi$

Sum of zeros = $10 = 10a \therefore a = 1$

Prod of zeros = $1460 = (4a^2 + b^2)(9a^2 + 4b^2)$

$\therefore (4 + b^2)(9 + 4b^2) = 1460$

$4b^4 + 25b^2 + 36 = 1460$

$4b^4 + 25b^2 - 1424 = 0$

$b^2 = \frac{-25 \pm \sqrt{625 + 16 \times 1424}}{8}$

$= \frac{-25 \pm 153}{8} = \frac{128}{8} \text{ or } -\frac{178}{8}$

$= 16 \text{ or } -22\frac{1}{4}$

but b integer

$\therefore b = \pm 4$

$\therefore$ Zeros: $2 \pm 4i$, $3 \pm 8i$

(ii) $\begin{matrix} \uparrow & \uparrow \\ \text{sum} = 4 & \text{sum} = 6 \\ \text{Prod} = 4 + 16 = 20 & \text{Prod} = 9 + 64 = 73 \end{matrix}$

$\therefore (z^2 - 4z + 20)(z^2 - 6z + 73)$

(c) $3^1 > 2(1) + \frac{1}{2} \ln 1$

$3^2 > 2(2) + \frac{1}{2} \ln 2$

$3^3 > 2(3) + \frac{1}{2} \ln 3$

$\vdots$
 $3^n > 2(n) + \frac{1}{2} \ln n$

$\therefore \underbrace{3^1 + 3^2 + 3^3 + \dots + 3^n}_{\text{LHS}} > 2(1+2+3+\dots+n) + \frac{1}{2}(\ln 1 + \ln 2 + \ln 3 + \dots + \ln n)$

LHS: $a=3, r=3, n=n$

$\therefore \frac{3(3^n - 1)}{3 - 1} > 2 \frac{n(n+1)}{2} + \frac{1}{2} (\ln(1 \times 2 \times 3 \times \dots \times n))$

$\therefore \frac{3}{2}(3^n - 1) > n(n+1) + \frac{1}{2} \ln(n!)$

$3(3^n - 1) > 2n^2 + 2n + \ln(n!)$

$3^{n+1} - 3 > 2n^2 + 2n + \ln(n!)$

$\therefore 3^{n+1} > 2n^2 + 2n + 3 + \ln(n!)$

(13) (a) $n=1: u_1 = \frac{1}{3}(2^2 + (-1)^1) = \frac{1}{3}(4-1) = \frac{3}{3} = 1$

$\therefore$ True for $n=1$

$n=2: u_2 = \frac{1}{3}(2^3 + (-1)^2) = \frac{1}{3}(8+1) = \frac{9}{3} = 3$

$\& u_2 = 2u_1 + (-1)^2 = 2(1) + 1 = 3$

$\therefore$ True for $n=2$

Let $n=k$ be true integer such that

$u_k = \frac{1}{3}(2^{k+1} + (-1)^k)$

Prove $n=k+1$ is integer such that

$u_{k+1} = \frac{1}{3}(2^{k+2} + (-1)^{k+1})$

Proof: LHS = $2u_k + (-1)^{k+1}$

$= \frac{2}{3}(2^{k+1} + (-1)^k) + (-1)^{k+1} \left[\text{from left statement} \right]$

$= \frac{1}{3}(2 \cdot 2^{k+1} + 2(-1)^k) + (-1)^1 (-1)^k$

$= \frac{1}{3}(2^{k+2} + 2(-1)^k + 3(-1)^1 (-1)^k)$

$= \frac{1}{3}(2^{k+2} + 2(-1)^k - 3(-1)^k)$

$= \frac{1}{3}(2^{k+2} - (-1)^k)$

$= \frac{1}{3}(2^{k+2} + (-1)(-1)^k)$

$= \frac{1}{3}(2^{k+2} + (-1)^{k+1}) = \text{RHS}$

$\therefore$ Proved by P.N.I

(b)(i) Let $u = a - x$

$\therefore \frac{du}{dx} = -1$

$\therefore dx = -du$

$x=0, u=a$

$x=a, u=0$

$\therefore \int_0^a f(a-x) dx = \int_a^0 f(u) (-du)$

$= -\int_a^0 f(u) du$

$= \int_0^a f(u) du$

$= \int_0^a f(x) dx$
by change of variable

3 (b) (ii)

$$I = \int_0^{\frac{\pi}{4}} \frac{\sin 2x}{\sin 2x + \cos 2x} dx = \int_0^{\frac{\pi}{4}} \frac{\sin \left[2 \left(\frac{\pi}{4} - x \right) \right]}{\sin \left[2 \left(\frac{\pi}{4} - x \right) \right] + \cos \left[2 \left(\frac{\pi}{4} - x \right) \right]} dx$$

$$I = \int_0^{\frac{\pi}{4}} \frac{\sin \left(\frac{\pi}{2} - 2x \right)}{\sin \left(\frac{\pi}{2} - 2x \right) + \cos \left(\frac{\pi}{2} - 2x \right)} dx$$

$$I = \int_0^{\frac{\pi}{4}} \frac{\cos 2x}{\cos 2x + \sin 2x} dx$$

$$\therefore I + I = \int_0^{\frac{\pi}{4}} \frac{\sin 2x + \cos 2x}{\cos 2x + \sin 2x} dx$$

$$2I = \int_0^{\frac{\pi}{4}} dx$$

$$I = \frac{1}{2} \left[x \right]_0^{\frac{\pi}{4}} = \frac{\pi}{8}$$

(c) (i) $\frac{d}{dx} \frac{1}{2} v^2 = 4x^3 - 12x$

$$\therefore \frac{1}{2} v^2 = x^4 - 6x^2 + c$$

When $x=0, v=0$

$$\therefore \frac{v^2}{2} = c$$

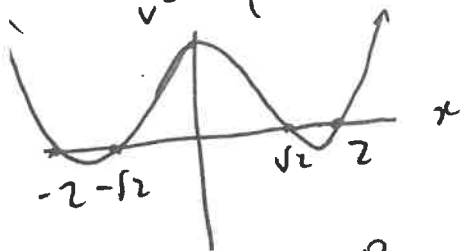
$$\therefore \frac{1}{2} v^2 = x^4 - 6x^2 + \frac{v^2}{2}$$

$$\therefore v^2 = 2x^4 - 12x^2 + v^2$$

(ii) $v^2 = 2x^4 - 12x^2 + 16$

$$= 2(x^4 - 6x^2 + 8)$$

$$v^2 = 2(x^2 - 4)(x^2 - 2)$$



Since initially $x=0$,
particle oscillates between $-\sqrt{2}$ & $\sqrt{2}$
since beyond these points, $v^2 < 0$
which is impossible.

At $x = \sqrt{2}, \dot{x} = -4/\sqrt{2}, v=0$
 $\therefore$ particle heads to the left

At $x = -\sqrt{2}, \dot{x} = 4/\sqrt{2}, v=0$
 $\therefore$ particle heads to the right

confirming
 $-\sqrt{2} \leq x \leq \sqrt{2}$

(iii) $\ddot{x} = -kx(3+x^2)$

$$\neq -n^2(x-a)$$

$\therefore$ Not SHM

(14) (a) (i) $M\ddot{x} = -\frac{1}{5}M$ at touchdown.

$$\therefore \ddot{x} = -0.2$$

$$\therefore 80 = c$$

$$\frac{dv}{dt} = -0.2$$

$$v = -0.2t + c$$

When $t=0, v=80$

$$\frac{dx}{dt} = 80 - 0.2t$$

$$\therefore x = 80t - 0.1t^2 + c$$

At $t=0, x=0 \therefore c=0$

$$\therefore x = 80t - 0.1t^2$$

At $t=10, x = 800 - 10$
 $= 790m$

(ii) For $t > 10$,

$$M\ddot{x} = -\frac{1}{5}M - \frac{1}{125}Mv^2$$

$$\therefore \ddot{x} = -\frac{1}{5} - \frac{1}{125}v^2$$

$$\therefore \frac{d^2x}{dt^2} = -\frac{1}{125}(25+v^2)$$

(iii) $v \frac{dv}{dx} = -\frac{1}{125}(25+v^2)$

$$\therefore \frac{dv}{dx} = -\frac{1}{125} \left(\frac{25+v^2}{v} \right)$$

$$\therefore \frac{dx}{dv} = -125 \left(\frac{v}{25+v^2} \right) = -\frac{125}{2} \left(\frac{2v}{25+v^2} \right)$$

$$\therefore x = -\frac{125}{2} \ln |25+v^2| + c$$

When $x=790, v=78$

$$\therefore 790 = -\frac{125}{2} \ln 6109 + c \therefore c = 790 + \frac{125}{2} \ln 6109$$

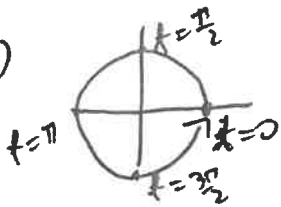
$$\therefore x = 790 + 62.5 \ln 6109 - 62.5 \ln (25+v^2)$$

$$x = 790 + 62.5 \left[\ln 6109 - \ln (25+v^2) \right]$$

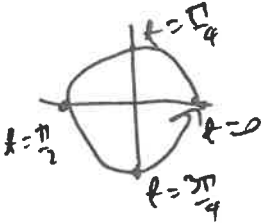
(iv) $v=0: x = 790 + 62.5 [\ln 6109 - \ln 25]$
 $= 1133.665(59 \dots m)$

$$\approx 1134m (1.m)$$

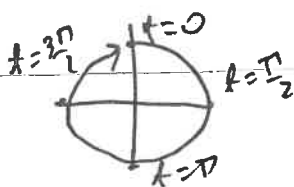
14(b)



$R_1(t)$: Circular
2π
anticlockwise



$R_2(t)$: Circular
π
anticlockwise



$R_3(t)$: Circular
2π
clockwise

$$\frac{1}{3\sqrt{2}} \tan^{-1} \frac{\sqrt{2}x}{3} = t + \frac{1}{3\sqrt{2}} \tan^{-1} \frac{\sqrt{2}}{3}$$

$$\tan^{-1} \frac{\sqrt{2}x}{3} = 3\sqrt{2}t + \tan^{-1} \frac{\sqrt{2}}{3}$$

$$\therefore \frac{\sqrt{2}x}{3} = \tan \left[3\sqrt{2}t + \tan^{-1} \frac{\sqrt{2}}{3} \right]$$

$$\therefore x = \frac{3}{\sqrt{2}} \tan \left[3\sqrt{2}t + \tan^{-1} \frac{\sqrt{2}}{3} \right]$$

OR: $t = \frac{1}{3\sqrt{2}} \left(\tan^{-1} \frac{\sqrt{2}x}{3} - \tan^{-1} \frac{\sqrt{2}}{3} \right)$

$$\tan(3\sqrt{2}t) = \frac{\frac{\sqrt{2}x}{3} - \frac{\sqrt{2}}{3}}{1 + \frac{\sqrt{2}x}{3} \cdot \frac{\sqrt{2}}{3}}$$

$$\left(1 + \frac{2x}{9}\right) \tan(3\sqrt{2}t) = \frac{\sqrt{2}x}{3} - \frac{\sqrt{2}}{3}$$

$$x \left[\frac{2}{9} \tan(3\sqrt{2}t) - \frac{\sqrt{2}}{3} \right] = -\frac{\sqrt{2}}{3} - \tan(3\sqrt{2}t)$$

$$x = \frac{\frac{\sqrt{2}}{3} + \tan(3\sqrt{2}t)}{\frac{\sqrt{2}}{3} - \frac{2}{9} \tan(3\sqrt{2}t)} = \frac{3\sqrt{2} + 9 \tan(3\sqrt{2}t)}{3\sqrt{2} - 2 \tan(3\sqrt{2}t)}$$

(c)

$$\ddot{x} = 8x^3 + 36x$$

$$\frac{1}{2}v^2 = 2x^4 + 18x^2 + c$$

At $x=1, v=11$

$$\therefore \frac{121}{2} = 2 + 18 + c$$

$$\therefore c = 40\frac{1}{2}$$

$$\therefore \frac{1}{2}v^2 = 2x^4 + 18x^2 + 40\frac{1}{2}$$

$$\therefore v^2 = 4x^4 + 36x^2 + 81$$

$$v^2 = (2x^2 + 9)^2$$

At $x=1, v=11$

$$\therefore v = 2x^2 + 9 \text{ not } -(2x^2 + 9)$$

$$\frac{dx}{dt} = 2x^2 + 9$$

$$\frac{dx}{2x^2 + 9} = \frac{1}{2} \times \frac{1}{x^2 + \frac{9}{2}}$$

$$\therefore t = \frac{1}{2} \frac{\sqrt{2}}{3} \tan^{-1} \frac{\sqrt{2}x}{3} = \frac{1}{3\sqrt{2}} \tan^{-1} \frac{\sqrt{2}x}{3} + c$$

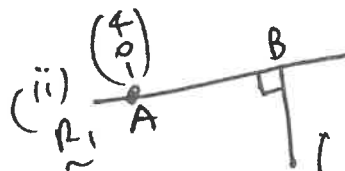
At $x=1, t=0 \therefore 0 = \frac{1}{3\sqrt{2}} \tan^{-1} \frac{\sqrt{2}}{3} + c$

$$\therefore c = -\frac{1}{3\sqrt{2}} \tan^{-1} \frac{\sqrt{2}}{3}$$

$$\therefore t = \frac{1}{3\sqrt{2}} \tan^{-1} \frac{\sqrt{2}x}{3} - \frac{1}{3\sqrt{2}} \tan^{-1} \frac{\sqrt{2}}{3}$$

(15) (a)(i) $\begin{pmatrix} 4 \\ \lambda \\ 1+\lambda \end{pmatrix} = \begin{pmatrix} 3+\mu \\ 3\mu \\ 2+2\mu \end{pmatrix}$ $\begin{matrix} \leftarrow \mu=1 \text{ at P.I} \\ \leftarrow \lambda=3 \\ \leftarrow \text{consistency!} \end{matrix}$

$\therefore$ Intercept at $\begin{pmatrix} 4 \\ 3 \\ 4 \end{pmatrix}$



$$\vec{AB} = \text{proj}_{\vec{AB}} \vec{AP} = \frac{(\vec{AP} \cdot \vec{AB})}{\vec{AB} \cdot \vec{AB}} \vec{AB}$$

$$= \frac{(-3) \cdot 0 + 4 \cdot 3 + (-1) \cdot 4}{(-3)^2 + 4^2 + (-1)^2} \vec{AB} = \frac{3}{2} \vec{AB}$$

$$\vec{PB} = \vec{PA} + \vec{AB} = \begin{pmatrix} -3 \\ 4 \\ -1 \end{pmatrix} + \frac{3}{2} \begin{pmatrix} 0 \\ 3 \\ 1 \end{pmatrix} = \begin{pmatrix} -3 \\ -2.5 \\ 2.5 \end{pmatrix}$$

$$\therefore |\vec{PB}| = \text{shortest distance} = \sqrt{9 + 6.25 + 6.25} = \sqrt{21.5} \text{ units}$$

Note: B is $\begin{pmatrix} 4 \\ 1.5 \end{pmatrix}$ & $\lambda = 1.5$ at B & also

15 (b) $I_n = \int_0^1 x \cdot \frac{1}{(1+x^2)^n} dx, n > 1$

(i) $I_n = \left[x \cdot \frac{1}{(1+x^2)^n} \right]_0^1 - \int_0^1 x(-n)(1+x^2)^{-n-1} (2x) dx$
 $= \left(\frac{1}{2^n} - 0 \right) + 2n \int_0^1 x^2 (1+x^2)^{-n-1} dx$
 $= \frac{1}{2^n} + 2n \int_0^1 \frac{x^2}{(1+x^2)^{n+1}} dx$

(ii) $I_n = \frac{1}{2^n} + 2n \int_0^1 \frac{(x^2+1)-1}{(1+x^2)^{n+1}} dx$
 $I_n = \frac{1}{2^n} + 2n \int_0^1 \frac{1}{(1+x^2)^n} - \frac{1}{(1+x^2)^{n+1}} dx$
 $I_n = \frac{1}{2^n} + 2n [I_n - I_{n+1}]$

$I_n = \frac{1}{2^n} + 2n I_n - 2n I_{n+1}$

$2n I_{n+1} = \frac{1}{2^n} + 2n I_n - I_n$

$2n I_{n+1} = \frac{1}{2^n} + I_n (2n-1)$

$I_{n+1} = \frac{1}{2n 2^n} + I_n \frac{(2n-1)}{2n}$

$I_{n+1} = \frac{1}{2^{n+1}} + \frac{2n-1}{2n} I_n$

(iii) Let $n=1$
 $I_2 = \frac{1}{1 \times 2^2} + \frac{1}{2} I_1 = \frac{1}{4} + \frac{1}{2} I_1$

$I_1 = \int_0^1 \frac{1}{1+x^2} dx = \left[\tan^{-1} x \right]_0^1 = \frac{\pi}{4}$

$\therefore I_2 = \frac{1}{4} + \frac{1}{2} \times \frac{\pi}{4} = \frac{\pi}{8} + \frac{1}{4}$

OR: $\vec{OF} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ $\vec{DF} = \begin{pmatrix} x-1 \\ y \\ z \end{pmatrix}$ $\vec{EF} = \begin{pmatrix} x-\frac{1}{2} \\ y-\frac{\sqrt{3}}{2} \\ z \end{pmatrix}$

$|\vec{OF}| = |\vec{DF}| = |\vec{EF}| = 1$

$\therefore 1 = x^2 + y^2 + z^2$
 $1 = (x-1)^2 + y^2 + z^2$

$(x-1)^2 = z^2$
 $1 = 2x$
 $x = \frac{1}{2}$

$\therefore \frac{3}{4} = y^2 + z^2$

But also $1 = (x-\frac{1}{2})^2 + (y-\frac{\sqrt{3}}{2})^2 + z^2$
 $x = \frac{1}{2} \therefore 1 = (y-\frac{\sqrt{3}}{2})^2 + z^2$

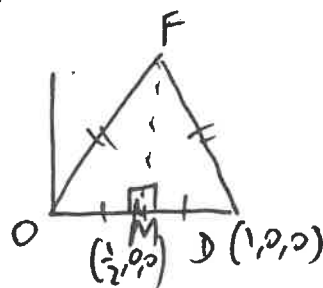
$\therefore -\sqrt{3}y + \frac{3}{4} = \frac{1}{4}$

$\sqrt{3}y = \frac{1}{2}$
 $\therefore y = \frac{1}{2\sqrt{3}} = \frac{\sqrt{3}}{6}$

$\therefore 1 + \frac{3}{4} + z^2 = \frac{1}{4} + \frac{3}{4}$

(c)

(i)



$|\vec{OF}| = 1$ $|\vec{OM}| = \frac{1}{2}$

$\therefore |\vec{MF}| = \sqrt{1 - \frac{1}{4}} = \frac{\sqrt{3}}{2}$

$\therefore E(\frac{1}{2}, \frac{\sqrt{3}}{2}, 0)$

(ii) Let $F(x, y, z)$
 $\cos \angle DOF = \frac{\vec{OF} \cdot \vec{OD}}{|\vec{OF}| |\vec{OD}|}$

$\cos \frac{\pi}{3} = \frac{\begin{pmatrix} x \\ y \\ z \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}}{1 \times 1}$

$\therefore \frac{1}{2} = x$

$\cos \angle EOF = \frac{\vec{OE} \cdot \vec{OF}}{|\vec{OE}| |\vec{OF}|}$

$\frac{1}{2} = \frac{\begin{pmatrix} \frac{1}{2} \\ \frac{\sqrt{3}}{2} \\ 0 \end{pmatrix} \cdot \begin{pmatrix} \frac{1}{2} \\ y \\ z \end{pmatrix}}{1 \times 1}$

$\therefore \frac{1}{2} = \frac{1}{4} + \frac{\sqrt{3}}{2} y \therefore \frac{1}{4} = \frac{\sqrt{3}}{2} y \therefore y = \frac{1}{2\sqrt{3}}$

$\therefore F(\frac{1}{2}, \frac{\sqrt{3}}{6}, \frac{1}{3})$

Now $|\vec{OF}| = 1 \therefore 1 = \frac{1}{4} + \frac{3}{36} + z^2$

$\therefore \frac{2}{3} = z^2 \therefore z = \frac{\sqrt{2}}{\sqrt{3}} = \frac{\sqrt{6}}{3}$

$\therefore F(\frac{1}{2}, \frac{\sqrt{3}}{6}, \frac{\sqrt{6}}{3})$

(16) Since $\sec^2 \theta - 1 = \tan^2 \theta$,
 (a) let $x = \frac{\sec \theta}{2} \therefore \frac{dx}{d\theta} = \frac{\sec \theta \tan \theta}{2}$

$$\therefore \int \frac{dx}{\sqrt{4x^2 - 1}} = \frac{1}{2} \int \frac{\sec \theta \tan \theta d\theta}{\sqrt{\sec^2 \theta - 1}}$$

$$= \frac{1}{2} \int \frac{\sec \theta \tan \theta d\theta}{\tan \theta}$$

$$= \frac{1}{2} \int \sec \theta d\theta$$

$$= \frac{1}{2} \int \frac{\sec \theta (\sec \theta + \tan \theta)}{\sec \theta + \tan \theta}$$

$$= \frac{1}{2} \ln |\sec \theta + \tan \theta|$$

Since $\sec \theta = 2x$,


$$= \frac{1}{2} \ln |2x + \sqrt{4x^2 - 1}| + C$$

(b) Note that divisibility by primes 2, 3 and 5 means divisible by $2 \times 3 \times 5 = 30$.

$$n=2: 2^5 - 2 = 32 - 2 = 30 = 30 \times 1$$

$\therefore$ True for $n=2$.

Let $n=k$ be integer ≥ 2 such that

$$k^5 - k = 30l \quad (l \text{ integer})$$

Prove $n=k+1$ is integer such that

$$(k+1)^5 - (k+1) = 30m \quad (m \text{ integer})$$

Proof: LHS = $(k+1)^5 - (k+1)$

$$= k^5 + 5k^4 + 10k^3 + 10k^2 + 5k + 1 - k - 1$$

$$= k^5 + 5k^4 + 10k^3 + 10k^2 + 4k$$

$$= 30l + 5k^4 + 10k^3 + 10k^2 + 5k$$

$$= 30l + 5(k^4 + 2k^3 + 2k^2 + k)$$

$$= 30l + 5k(k^3 + 2k^2 + 2k + 1)$$

$$= 30l + 5k(k+1)(k^2 + k + 1)$$

We'd like $6 = 3! \text{ as a factor}$
 $\therefore$ we'd like 3 consecutive numbers

$$= 30l + 5k(k+1)((k-1)^2 + 3k)$$

$$= 30l + 5k(k+1)(k-1)^2 + 15k^2(k+1)$$

$$= 30l + 5(k-1)(k-1)k(k+1) + 15k \frac{k(k+1)}{2}$$

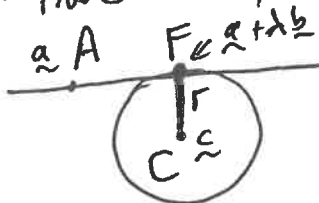
$= 6p, p \in \mathbb{Z} \quad = 2q, q \in \mathbb{Z}$

$$= 30l + 30p(k-1) + 30qk$$

$$= 30(l + p(k-1) + qk) = 30m$$

Since l, p, k, q integers.

$\therefore$ Prove true by P.M.I

(c)(i) 

$$\frac{AF}{AC} = \frac{PF}{FC} = \frac{r}{b} \Rightarrow \frac{AF}{AC} = \frac{r}{b} \Rightarrow AF = \frac{r \cdot AC}{b}$$

$$|a + \lambda b - c| = r$$

$$\therefore r^2 = [(a-c) + \lambda b] \cdot [(a-c) + \lambda b]$$

$$r^2 = (a-c) \cdot (a-c) + 2\lambda b \cdot (a-c) + \lambda^2 b \cdot b$$

$$\lambda^2 |b|^2 + 2\lambda b \cdot (a-c) + |a-c|^2 - r^2 = 0$$

$$\Delta = [2b \cdot (a-c)]^2 - 4|b|^2 [|a-c|^2 - r^2]$$

$$= 4[b \cdot (a-c)]^2 - 4|b|^2 [|a-c|^2 - r^2]$$

Since tangent, λ has only one value $\Delta = 0$

$$\therefore [b \cdot (a-c)]^2 = |b|^2 [|a-c|^2 - r^2]$$

$$\therefore \lambda^2 |b|^2 + 2\lambda b \cdot (a-c) + [b \cdot (a-c)]^2 = 0$$

$$\therefore \left[\lambda |b| + \frac{b \cdot (a-c)}{|b|} \right]^2 = 0$$

$$\therefore \lambda = \frac{b \cdot (c-a)}{|b||b|} = \frac{b \cdot (c-a)}{|b|^2}$$

16(c)(ii)

$$\vec{CF} = \vec{a} + \lambda \vec{b} - \vec{c}$$

$$\text{Now } \vec{CF} \cdot \vec{b} = (\vec{a} + \lambda \vec{b} - \vec{c}) \cdot \vec{b}$$

direction
vector

$$= \vec{a} \cdot \vec{b} - \vec{b} \cdot \vec{c} + \lambda \vec{b} \cdot \vec{b}$$

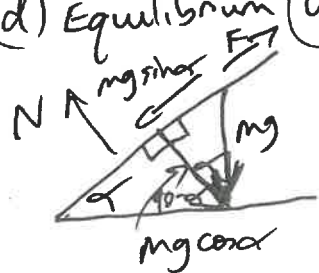
$$= \vec{a} \cdot \vec{b} - \vec{b} \cdot \vec{c} + \lambda |\vec{b}|^2$$

$$\text{Using (i)} \\ = \vec{a} \cdot \vec{b} - \vec{b} \cdot \vec{c} + \frac{\vec{b} \cdot (\vec{c} - \vec{a})}{|\vec{b}|^2} |\vec{b}|^2$$

$$= \vec{a} \cdot \vec{b} - \vec{b} \cdot \vec{c} + \vec{b} \cdot \vec{c} - \vec{b} \cdot \vec{a} = 0$$

$\therefore$ Perpendicular!

(d) Equilibrium (with angle α):



$$\therefore F = mg \sin \alpha$$

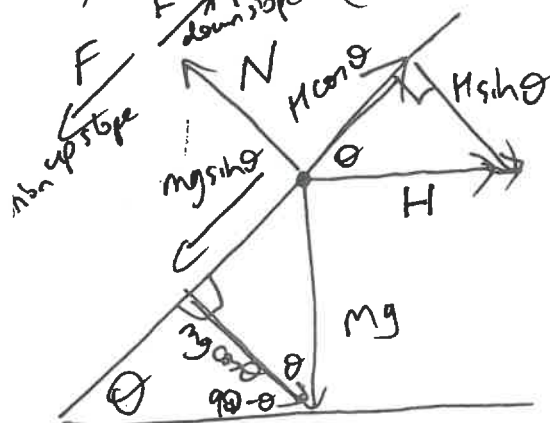
$$N = mg \cos \alpha$$

$$\therefore \frac{F}{N} = \tan \alpha$$

$$\therefore \frac{MN}{N} = \tan \alpha$$

$$\therefore \mu = \tan \alpha$$

After H applied (with angle θ)



$$\text{So: } N = H \sin \theta + mg \cos \theta$$

$$\text{Motion up slope: } H \cos \theta > mg \sin \theta + F$$

$$\text{Motion down slope: } mg \sin \theta > H \cos \theta + F$$

$\therefore$ For motion up slope not to happen,

$$H \cos \theta \leq mg \sin \theta + F$$

& for motion down slope not to happen,

$$mg \sin \theta \leq H \cos \theta + F$$

$$\rightarrow H \cos \theta \leq mg \sin \theta + \mu N$$

$$H \cos \theta \leq mg \sin \theta + \tan \alpha (H \sin \theta + mg \cos \theta)$$

To create $\tan$, $\div \cos$

$$\therefore H \leq mg \tan \alpha + \tan \alpha (H \tan \theta + mg)$$

$$H(1 - \tan \theta \tan \alpha) \leq mg \tan \alpha + mg \tan \alpha$$

$$H \leq \frac{mg(\tan \alpha + \tan \alpha)}{1 - \tan \theta \tan \alpha}$$

$$H \leq mg \tan(\theta + \alpha)$$

$$\rightarrow mg \sin \theta \leq H \cos \theta + \tan \alpha (H \sin \theta + mg \cos \theta)$$

$\div \cos$

$$mg \tan \theta \leq H + \tan \alpha (H \tan \theta + mg)$$

$$mg \tan \theta - mg \tan \alpha \leq H + H \tan \theta \tan \alpha$$

$$mg(\tan \theta - \tan \alpha) \leq H(1 + \tan \theta \tan \alpha)$$

$$\frac{mg(\tan \theta - \tan \alpha)}{1 + \tan \theta \tan \alpha} \leq H$$

$$mg \tan(\theta - \alpha) \leq H$$

$$\therefore mg \tan(\theta - \alpha) \leq H \leq mg \tan(\theta + \alpha)$$